What limits any system that improves itself, whatever it is made of?

The founding paper of Recursive Dynamics: the field as a whole, with the five laws of the ARC Theory as laws under test

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Abstract

Machines that improve themselves using their own outputs now exist and work, and there is no accepted science of what limits them. This paper founds that science, Recursive Dynamics, and stakes its claim the way thermodynamics was staked in 1824: by computing a bound that any such machine must obey if the claim is right, publishing the instruments that measure its terms, and preregistering by timestamp the experiments whose nulls favour the rival view. The field takes the self-improvement loop as its native object and names five state variables, conjectured to be substrate-independent: capability, recursive depth, correction rate, drift and correction leverage. It carries five laws as named conjectures under test, with statuses declared. One, the ARC Ceiling, is derived in full from two lines of assumption; two, added on 24 September 2026 and registered as laws on 15 October 2026 in version 1.103 of the theory registration, are not yet tested, and their mathematics, set out at version 1.1, is in development. The field locates its identity, as thermodynamics does, in its impossibilities: if the laws hold, self-improvement cannot stably outrun its own correction, the growth exponent is capped by the reciprocal of the correction shortfall, and no ladder of same-class oversight climbs past its ceiling by adding rungs. The paper states what would make the field wrong, tries to house its questions inside twelve neighbouring fields, gives each of fifty-one objections a disposition, and prints the four results that would dissolve the field. Its honest status is the Carnot stage: one memoir, instruments built, decisive measurements preregistered by timestamp and not run, staked where failure will be as visible as success.

Version 3.5 sets the paper in this form, states Laws IV and V as registered on 15 October 2026 and makes the corrections recorded for it, among them Law II's one-line statement, the premise that fixes the ceiling's value and the bounds that prior work states, now conceded exactly; Appendix A lists every version and correction, with its date.

The founding paper of Recursive Dynamics, the field as a whole: it states the five laws of the ARC Theory, the field's founding theory, as named conjectures under test (Laws I to III registered on 8 September 2026; Laws IV and V added on 24 September 2026 and registered as laws on 15 October 2026 in version 1.103 of the theory registration, not yet tested, their mathematics in development) and derives Law III, the ARC Ceiling, inside a stated model; Appendix D gives its place in the programme. Working paper, version 3.5, 24 October 2026; first published 31 August 2026; Appendix A lists every version and correction. Recorded on OSF as *Founding Paper · Recursive Dynamics · The Science of Systems That Improve Themselves*, DOI 10.17605/OSF.IO/HCPBU. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; Paper C, on polymathy: doi.org/10.17605/OSF.IO/RBHDP. Licence: CC BY 4.0.

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1 Introduction

In 1824 the steam engine was the most consequential machine on earth, and nobody could say what limited it. Engineers improved engines the way engineers do, by building the next one; the question of what any engine could do, independent of its construction, was not merely unanswered but unasked, because it belonged to no existing subject. Sadi Carnot asked it in a memoir (Carnot 1824) that founded a science, and the pattern has repeated ever since: cybernetics was proposed as a field in 1948, in the book that named it (Wiener 1948), before most of its results existed; exobiology was named in 1960 (Lederberg 1960) before a single specimen. When working machines precede their theory, the theory that forms is not a branch of an old field. It is a new one, because the old fields' native objects were never the machine's behaviour as such. The present moment has exactly this shape. Systems now improve their own outputs using their own outputs: models that critique and revise their reasoning, pipelines that generate their own training signal, laboratories automating parts of their own research loop (Chen et al. 2026). They work. Whether they stay under control as they grow is the control problem of AI safety (Bostrom 2014), and for systems that improve themselves it is a question of rates: whether their correction keeps pace with their growth. Ask what bounds them and the literature answers with scenarios, forecasts and intuitions, and with formal models, bounds and fitted scaling laws, each stated in its own variables (Section 2); it has not yet answered with one set of state variables in which a limit can be stated once and tested across substrates. That is not a criticism of the literature; it is the diagnostic signature of a missing field.

This paper founds that field. Its one derived result is a ceiling: inside a stated minimal model, correction out-scales the burden that growth creates only while the growth exponent stays below a value set by how correction capacity scales with capability, and for a corrector of the system's own class that value is at most two, on a premise the programme names as the conjecture's most likely point of failure (Section 6). It is written to be attacked. Section 7 makes the positive case as a scored list of conditions; Section 12 collects every objection its author and his adversarial readers could find and prints what happened to each; Section 14 states what would dissolve the field; Section 15.1 states what would have to happen before anyone should call it a science.

Our contributions are as follows.

1. The field's object and its measurement principle: the self-improvement loop as the native object, which none of nine neighbouring fields takes as primitive (Section 3), and capability measured on a ratio scale with a true zero and no ceiling, with five state variables written as the programme's notation register defines them (Section 4).
2. The five laws of the ARC Theory stated as named conjectures under test, each with its status (Section 5), and one bound worked in full: inside a stated minimal model the burden-to-capacity ratio scales as $R^{\alpha(1-\gamma)-1}$, so the crossover is $\alpha_{\text{crit}} = 1/(1-\gamma)$, and at the same-class premise $\gamma = 1/2$ it is 2 (Section 6).
3. The claim stated as what it forbids, four outcomes under the preregistered analyses (Section 1.1), and the four results that would dissolve the field, with the commitment to publish each under this title (Section 14).
4. An attempt to house the field's questions in twelve candidate host fields in their own variables, conceding what each already holds, with a dissolution map by host and the one coupling no host can state (Section 3.1).
5. The case for the field scored against six licence conditions, with the precedent class of fields founded on a limit theorem and its disanalogy (Section 7); the rival's best case, with the programme's own contrary pilot printed (Section 8); the grade of substrate independence (Section 9); the instruments (Section 10); and the decisive experiments for Laws I to III, preregistered by timestamp and not run, with what would confirm a new area rather than a new chapter (Section 11).

6. Fifty-one objections to founding the field, each stated in its strongest form with a printed disposition (Section 12).

Section 2 sets out, by name, the debts the field owes and the later work beside it. Section 3 defines the object and tries twelve hosts for it; Section 4 gives the measurement principle and the five state variables; Section 5 states the five laws with their statuses; and Section 6 works one bound in full. Sections 7 to 11 make the case for the field, give the rival's best case, grade substrate independence and set out the instruments and the decisive experiments. Section 12 gives every objection with its disposition, Section 13 what the field buys if it holds and if it fails, and Section 14 what would dissolve it. Section 15 states what the paper does and does not show, Section 16 its limitations, and Section 17 concludes. Appendix A gives the version history and every correction, Appendix B the priority record, Appendix C the notation and a glossary, and Appendix D the paper's place in the programme.

1.1 What is claimed, stated as what is forbidden

A claim that forbids nothing is not a claim, and a paper that concedes as much as this one does owes the reader its prohibitions first. Under the preregistered analyses, the field forbids four outcomes:

1. A stable system holding its growth exponent above the ceiling, $\alpha_{\text{crit}} = 1/(1-\gamma)$, for a sustained run.
2. A same-class corrector whose correction leverage γ measures wholly above one half.
3. A corrector-class contrast that runs the other way, or that falls wholly below the preregistered support threshold wherever the cap should bind: the decisive trial's own anti-directional and equivalence verdicts.
4. Monotone returns in the growth exponent, with no interior maximum, as reinvestment rises.

The first, the third and the fourth would each make the field wrong. The second is a prohibition of a different weight: it kills the one-half cap and the value two that follows from it, while the ceiling relation stands, because the relation returns whatever leverage exponent is measured. A true null on the corrector-class trial's six-family roster returns an inconclusive verdict at the preregistered precision rather than an equivalence verdict, which is reported as a failed attempt to confirm and never as a refutation. Everything conceded in Section 12 is conceded in the light of these four; none of the concessions softens them.

2 Related work: debts conceded by name

This field is assembled from debts, each conceded in full: the cybernetic intuition that a regulator must match the variety of what it regulates, so that control capacity must scale with the controlled system (Ashby 1956; Conant and Ashby 1970), to which this field adds the exponent form and the class lever; the impossibility line of AI-control research (Yampolskiy 2020), to which it adds the design requirement the impossibility implies; the transport-network limits of biological scaling (West et al. 1997), conceded as premise, to which it adds the internal replacement limit that operates when fixing the pipe no longer fixes the growth; the intelligence-explosion question as posed by its named owners from 1965 onward (Good 1966; Vinge 1993; Chalmers 2010; Hutter 2012; Bostrom 2014), to which it adds the missing state variables; the self-modifying-program line of the 2000s, whose Gödel machines rewrite any part of their own code once they have found a proof that the rewrite is useful (Schmidhuber 2003), the theoretical ancestry of a program that rewrites itself, to which it adds measured rates, an exponent and a class cap; the functional-equation mathematics, the machinery, never claimed; and the scalable-oversight programme (Irving et al. 2018; Christiano et al. 2018; Leike et al. 2018; Burns et al. 2023), whose ladders are the object the third law is about. What is claimed as original is the assembly: the loop as native object, scale

discipline as founding method, the bounds as laws, the corrector's class as the lever, and the instruments, preregistered by timestamp, that decide Laws I to III.

Formal treatments of recursive self-improvement predate this field, and this paper claims no priority for a formal treatment as such. Wang (2018) formally defines one class of recursively self-improving systems, proves that, in a restricted setting, such systems can be computable, and reports simulations suggesting that they can be efficient; Reddy (2020) sets out Markov-chain, history-embedding and program-generating models of recursive self-improvement. Chen et al. (2026) survey recursive self-improvement across 1,250 papers and list, among their open problems, stability engineering as a discipline and measurement fit for governance; that is context for the need this field answers, not evidence for any of its laws. On recursive error, models collapse when trained on recursively generated data (Shumailov et al. 2024), and in an analytically tractable linear setting the test error has a finite bound, independent of the number of iterations, when real and generated data accumulate rather than replace each other (Gerstgrasser et al. 2024). On correctors, Engels et al. (2025) fit scaling laws for the success of oversight under a capability gap between overseer and overseen and analyse nested scalable oversight, a quantitative baseline that any claim about how correction scales must beat; Rudoler et al. (2026) prove that, in a supervised setting, monitoring with vanishing error and query rates is possible exactly when the class of possible monitor functions has finite Littlestone dimension, and report results suggesting that monitors co-trained with adapting workers keep pace better than fixed monitors, in code-security settings; and Wang et al. (2026) report that, in the setting they study, self-evolution without execution-based verification fails to improve. Jafari et al. (2025) develop an analytic framework for recursive self-improvement in which limits from power, bandwidth and memory define a service envelope that caps instantaneous improvement, and an endogenous growth model coupling capital to compute, data and energy defines a critical boundary between superlinear and subcritical regimes; under the paper's reading rule, its comparison with this field waits until it has been read in full.

The neighbouring fields that could hold the field's questions are tried in their own variables in Section 3.1, with what each already holds conceded; the precedent class of fields founded on a limit theorem is set out in Section 7.1; and the objections that the field already exists under another name are answered in Section 12, group A.

3 The object, and why it belongs to no one

Definition 1 (Recursive system)**.** Define a recursive system as any system whose outputs re-enter its own improvement process.

The field's native object is that loop. Its neighbours each own a tool and none owns the object, and each is named here with what it does own, because the discipline is assembled from debts to them, conceded in Section 2.

A field is licensed when the questions that matter are about an object none of the existing fields takes as primitive. The questions that matter here, how fast can self-improvement go, what bounds it, and what kind of overseer can keep up, are questions about the loop.

3.1 The honest attempt to house it elsewhere

Table 1 is the kind of demarcation that reads as marketing to exactly the readers who matter, because it never tried the fit. This section tries. For each candidate host the three questions (how fast can self-improvement go; what bounds it; what overseer keeps up) are stated in the host's own variables, everything the host already covers is conceded to it, what breaks is stated precisely, a verdict is given, and the result that would move the field into that host is named. Twelve hosts, in rough order of how much of

Table 1. Neighbouring fields: what each owns, and why it does not own the loop.

Neighbouring field	What it owns	Why it does not own the loop
Dynamical systems theory	Change in general, fixed points, stability; it lends this field its mathematics	It does not take self-rewriting update rules as its object
Cybernetics	Regulation: a controller holding a plant to a reference; the requisite-variety intuition that control capacity must match what it controls (Ashby 1956)	It centres designed, external loops; here the loop rewrites the controller's own capability
Control theory	Stabilising a plant with a controller outside it	Here the controller is the plant, made of the same stuff, sharing its failure modes
Complexity science	Emergence and self-organisation	It carries no correction budget
Learning theory	The learner fixed, the data varied	Here the learner varies itself
Evolutionary dynamics	Variation and selection	No designer directs reinvestment into the improvement process itself
Scaling-law empirics	Capability against external resources (parameters, data, compute)	Here the resource of interest is the system's own re-entering output
Scalable oversight	Correctors, and ladders of them; bounds on oversight in its own variables	None of its bounds depends on what the corrector is made of, which is what the third law is about
Functional-equation mathematics	The machinery of the scaling forms	Claimed here as nothing more than machinery

the field each could hold. Where a host's statement rests on a canonical work, that work is cited for what is stated there; finer literatures not yet read are left uncited, under the paper's rule that nothing is cited before it is read.

1. Adaptive control and self-tuning regulation. Stated there: a plant with a controller whose gains adapt online (Åström and Wittenmark 1973); stability by a Lyapunov argument; the adaptation gain against the disturbance rate. What fits: Law II is, in form, an adaptive-control stability condition, one rate against another, and the loop that rewrites its own controller is half present, because a self-tuning regulator does rewrite its gains. What breaks: adaptive control keeps the controller's structure fixed and adapts parameters; it has no variable for the controller's own capability growing with depth, and no notion that a controller built from the plant's own class is capped in what it can correct; its conservation law (feedback cannot reduce sensitivity everywhere) is a law of fixed loops. Verdict: could host Law II as a chapter, "stability of self-modifying regulators", if the class variable proves empty. Moves there if: the corrector-class ratio holds at 1.00.

2. Queueing theory and the stability of service systems. Stated there: correctable burden arrives at a rate proportional to capability gained; correction is a server whose capacity scales with capability; the system is stable when the load ratio stays below one (Lindley 1952). What fits: this is the ARC Ceiling's derivation, and the paper concedes it. Section 6 is a queueing stability condition with an unusual coupling: the arrival rate grows with the work already served. What breaks: queueing theory has no notion that the server is built from the same class as the arrivals and therefore capped in how its capacity scales; it carries no depth exponent and assumes stationarity. Verdict: the strongest host for the *form* of Law III; if the class cap is empty, the ARC Ceiling belongs here as a growth-coupled queue and the field's name attaches to nothing in it. Moves there if: the eclipse measurement retires the class distinction.

3. Endogenous growth theory in economics. Stated there: a knowledge production function in which the rate of new knowledge depends on the existing stock raised to an exponent; the share of output reinvested in research is the policy dial; the field's long argument over whether that exponent is below one (growth needs ever more inputs) or at one and above (growth can be explosive) (Romer 1990; Jones 1995) is exactly the question of whether recursion is sustainable. What fits: Law I, taken alone, is a knowledge-production exponent under another name, and the reinvestment dial the preregistered studies turn is the savings rate of a growth model; the paper concedes both. What breaks: growth theory has no drift ledger, because ideas in its models do not corrupt the stock that produced them, and no corrector; it therefore cannot state Laws II or III, and it does not ask what the improver is made of. Verdict: could host Law I entirely. The field's claim to Law I is only through its coupling to Laws II and III; that coupling is the whole of what is claimed. Moves there if: drift and correction prove inseparable from capability under measurement (objection G3), leaving a growth exponent and nothing else.

4. Software reliability engineering and defect dynamics. Stated there: faults injected per change against faults repaired per unit effort; reliability growth when repair outpaces injection (Musa 1975); the accumulation of unrepaired debt. What fits: drift and the correction rate are, operationally, injection and repair rates, and the field's estimators for them owe this discipline their shape; conceded. What breaks: a reliability model does not let the system's repair capacity grow as a function of its own capability, has no class variable and no depth exponent. Verdict: could host the measurement methodology for the two rates, not the laws. Moves there if: the rates turn out to be measurable only as engineering metrics with no lawful relation between them.

5. Machine learning: self-play, self-training, learned optimisers and model collapse. Stated there: outputs re-entering training is the self-play loop; degradation of models trained on their own outputs is drift measured in special cases (Shumailov et al. 2024); learned optimisers are learners that rewrite the learner. What fits: every preregistered experiment could be published here as a machine-learning result, and this is the most likely institutional home if the field fails to earn its name; conceded without reservation. What breaks: the literature carries no ratio-scaled capability law across depth with a stated exponent, no bound on what a learner-rewriting learner can do, and no class cap. Its model-collapse line does pair drift with a correction, real data kept or accumulated beside generated data, and shows the bounded-versus-divergent dichotomy for training dynamics (Section 2); it does not vary what the corrector is made of. Verdict: hosts the experiments; cannot state the laws. Moves there if: the substrate crossing fails, leaving the laws true of language-model loops only, which is machine learning (objection H5).

6. Scaling-law empirics. Stated there: capability against parameters, data and compute; exponents fitted (Kaplan et al. 2020), architecture-blind by construction. What fits: the exponent-fitting craft, the model families, the discipline of reporting intervals; conceded. What breaks: the resource here is the system's own re-entering output; resource scaling predicts monotone returns where the field predicts an interior maximum. Verdict: hosts the method, not the object. Moves there if: recursive depth reduces, under measurement, to compute spent, or the titration shows monotone returns.

7. Scalable oversight and alignment research. Stated there: a weaker or same-class corrector supervising a stronger system (Burns et al. 2023); ladders of correctors (Engels et al. 2025); the premise that they scale. What fits: the corrector-class question is institutionally theirs, and the eclipse result, whichever way it falls, is an oversight result; conceded. What breaks: the literature bounds what a corrector can do in its own variables, the capability gap between overseer and overseen (Engels et al. 2025) and the complexity of the class of monitor functions (Rudoler et al. 2026), but none of its bounds

depends on whether the corrector shares the corrected system's architecture, training distribution and blind spots, so it cannot state the third law or the cap that its own ladders assume away. Verdict: the strongest institutional host for the class-cap result. If the field dissolves, the eclipse measurement is published here and belongs here. Moves there if: the ratio and the cap prove real but nothing couples them to a growth exponent in depth, which would make the finding an oversight theorem rather than a field.

8. Evolutionary computation and error-threshold theory. Stated there: strategy parameters that evolve alongside the solution (the improver improving itself, which self-adaptive evolution strategies have done for decades, Beyer and Schwefel 2002); and, in the theory of replicating populations, the error threshold, above which mutation outruns selection and information is lost (Eigen 1971). What fits: the error threshold is the closest existing law to Law II, a rate of corruption against a rate of correction with a boundary between persistence and collapse, and the paper concedes it as an antecedent of the second law's shape; self-adaptation is a working instance of the loop rewriting its improver. What breaks: selection is external to the replicator, the corrector has no class, and capability is not a ratio ladder. Verdict: could host Law II as a generalised error threshold. Moves there if: the correction-versus-drift race finds the boundary but the class lever does nothing.

9. Statistical physics: scaling, universality and renormalisation. Stated there: the same exponent families across unrelated substrates as a universality class (Wilson 1971); scaling collapse as the test. What fits: the field's substrate-independence hope is this tradition's hope, and its fourth-cell test is a universality test; conceded. What breaks: no correction variable, no drift variable, no corrector whose composition is the lever. Verdict: hosts the cross-domain families if they survive and belong to no loop. Moves there if: the families hold and the loop-specific laws do not.

10. Computability, algorithmic information and the self-referential improver line. Stated there: programs that rewrite any part of their own code once they have found a proof that the rewrite is useful, such a rewrite being globally optimal in the sense the proof fixes (Schmidhuber 2003); the limits of self-reference (Turing 1936). What fits: this is the theoretical ancestry of "a program that rewrites itself", conceded. What breaks: those results concern provability and optimality, not measured rates; they carry no exponent, no drift and no class cap. Verdict: ancestry, not a host. Moves there if: nothing empirical survives and only the self-reference question remains.

11. Second-order cybernetics, autopoiesis and general systems theory. Stated there: systems that produce and maintain themselves (Maturana and Varela 1980); the observer inside the loop. What fits: the conceptual picture of a self-producing system is theirs and is conceded. What breaks: the tradition is qualitative; it has no ratio ladder, no exponents, no preregistered decider. Verdict: a philosophical host, not a measuring one. Moves there if: the field's quantities prove unmeasurable, at which point it was never a science and belongs with the qualitative traditions of self-organisation.

12. Metascience and organisational learning. Stated there: science itself as the original self-correcting recursive improver; institutions that learn. What fits: an application domain and a candidate non-artificial substrate for the crossing. What breaks: no capability ladder, and the corrector is a community, not a class. Verdict: a domain the field would be tested in, not a home.

The dissolution map by host. If the founding conjunction fails, the parts do not vanish; they go home, and the paper says where. Law I returns to endogenous growth theory as a knowledge-production exponent. Law II returns to adaptive control and error-threshold theory as a rate condition. The ARC Ceiling returns

to queueing theory as a growth-coupled stability condition. The class-cap result, whichever way it falls, is published in scalable oversight. The instruments and the experiments are machine learning. The cross-domain families, if they survive, are statistical physics. Laws IV and V, added on 24 September 2026 and registered as laws on 15 October 2026, are not yet placed on this map. Nothing placed on it is orphaned by the field's death, which is what makes the death publishable.

The residue no host holds. After the twelve attempts, one thing remains that none of them can state in its own variables: a cap on correction leverage that depends on what the corrector is made of, coupled to a growth exponent in recursive depth, so that the sustainable rate of self-improvement is a property of the corrector's class. Growth theory has the exponent and no corrector; control and queueing have the stability condition and no class; oversight has the corrector and bounds in its own variables, and no cap set by the corrector's class; machine learning has the loop and conditions under which retraining on a model's own outputs stays bounded, and no class lever. Recursive Dynamics is the name for that coupling. If the coupling is empty, the parts go home and the name dies, and the field will have cost its neighbours nothing but a careful table.

4 Scale discipline, and the five state variables

Before any law, a measurement principle, learned the hard way and adopted as founding method: **capability must be measured on a ratio scale with a true zero and no ceiling.** Thermodynamics could not state its laws in their quantitative form until temperature was absolute; a bounded percentage score does to Recursive Dynamics what a Celsius-only thermometer does to the third law of thermodynamics: it makes the laws unstatable, because a score capped at 100 per cent cannot follow a power law in anything. The field's instrument is therefore a calibrated difficulty ladder: a latent, ratio-scaled capability measure with a true zero and no ceiling, on which doubling means the same thing at every height; and every quantitative surface names which of three model families it is reading (unbounded or latent capability, error decay, or bounded performance), because the exponent means something different in each. The programme adopted this after catching its own worked example running a bounded score through unbounded arithmetic; the correction is recorded, and the lesson is promoted here from a bug fix to a founding principle: **scale discipline precedes law.**

Five variables are conjectured to matter regardless of substrate, written exactly as the programme's notation register defines them, because the register is the one place a symbol is allowed to mean something:

Table 2. The five state variables, written as the programme's notation register defines them.

Variable	Symbol	What it is
Capability	U (or C)	What the system can do, on a calibrated ladder with no ceiling
Recursive depth	R	How many times outputs have re-entered the loop
Correction rate	β_C	How fast internal correction strengthens as capability grows; the quantity Law II asks to out-scale drift
Drift	k	How fast the errors and misalignments that accumulate per self-modification grow as capability grows; the exponent Law II asks the correction rate to out-scale
Correction leverage	γ	How correction capacity scales with capability; the exponent in the ARC Ceiling

Two of the five, capability and recursive depth, are states; the other three, the correction rate, drift and correction leverage, are elasticities of relations among them (local log-derivatives, which are exponents in

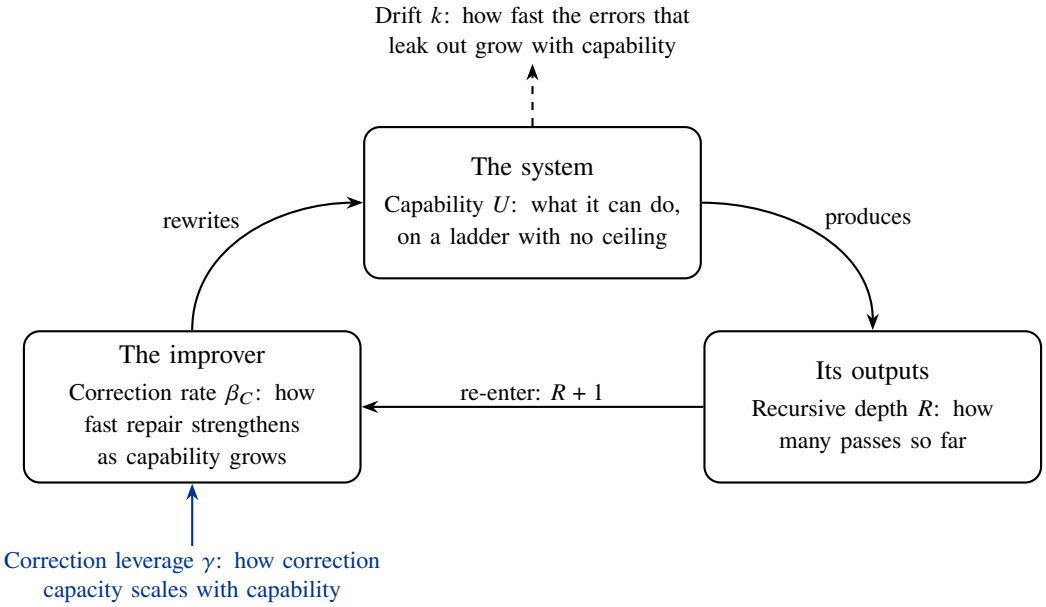


Figure 1. The loop, and the five quantities placed on it. Solid black: the loop; dashed: what leaks out; accent colour: what is fed back in. Five laws relate these quantities; each carries its status: a named conjecture, not established. Nothing in the picture is measured yet; it shows what the discipline sets out to measure, and what would refute the shortlist: if these five cannot state relations that hold under measurement, the shortlist is wrong.

the power-law case), each estimated over a range rather than read at an instant.

The share of effort reinvested in the improvement process itself is the experimental dial the preregistered studies turn: a manipulation, not a state variable, and it carries no symbol here on purpose, because the notation register records a bare β doing five different jobs across the corpus as a named error class.

The shortlist of five will look arbitrary. So did pressure, volume, temperature, energy and entropy, until the laws relating them held. The claim that five suffice, and that these are the five, is part of what the field must prove; the paper returns to it as objection C1.

5 The five laws, statuses declared

The field's laws are named conjectures, and their statuses travel with them everywhere, including here. "Law" is the field's naming convention for a relation it puts to the test, never a claim that the test has been passed.

Law I, the ARC Principle: $U = I \times R^{\alpha}$. Capability grows as a power of recursive depth from I, what the system can do at depth one. Status: a book-framework claim (in print 2 January 2026, Eastwood 2026a; framework dated 8 December 2024), with exploratory cross-domain support and its own early headline exponent publicly retracted: the 2.24 estimate appears in this programme only as a retraction, and its blinded replacement is a conversion measurement, an α estimate and never a γ, on one model: a regression estimate near 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, beside an endpoint estimate of 0.59 whose bootstrap 95 per cent interval, minus 1.3 to 2.9, is wide enough to include zero and two, and a compute-matched parallel exponent of 0.31. The form is under registered test: in the theory registration of 8 September 2026 the register ARC-1 tests the first law and holds P1, P2, P8, P9 and P18.

Law II, the ARC Co-Scaling Law: $\beta_C > k$. In the words of the theory registration, "correction has the asymptotic scaling advantage where its exponent exceeds the drift exponent, burden has it where the correction exponent is the smaller, and at equality coefficients, delay, backlog and saturation decide the outcome". Status: registered on 8 September 2026, with ARC-2 the register that tests it; a law under test, proved inside a stated minimal model and nowhere else; not established as necessary, not as sufficient, not yet measured on a system that genuinely rewrites itself. Its decider is the correction-versus-drift race, the field's simplest and most consequential measurement: a system whose correction durably out-scales its own drift is this field's Joule experiment.

Law III, the ARC Ceiling: $\alpha_{\text{crit}} = 1/(1-\gamma)$. The growth exponent is capped by the reciprocal of the correction shortfall. Status: registered on 8 September 2026, with ARC-3 the register that tests it; a law under test. Unlike the first two, this one can be derived in front of the reader, and is, in the next section. Its load-bearing premise, that γ is capped at one half for a corrector of the same class as its system, is the independence-aggregation benchmark, named by the programme itself as the conjecture's single most likely point of failure. A criticism of the derivation's depth treatment, raised in review, is printed as a limitation, not argued away; and the retired form of this law, with γ alone in the denominator, was retracted on 16 August 2026 and appears in this programme only as a retraction.

Law IV, the ARC Persistence Law: value persistence. Whether externally installed gains decay under subsequent capability training. Status: added to the theory on 24 September 2026; registered as a law on 15 October 2026 in version 1.103 and not yet tested; its mathematics is set out at version 1.1 of 7 October 2026 and is in development. One proposition of the registration of 8 September 2026 carries it, P15: more than half of an externally installed gain is lost once later capability training matches the budget that installed it. In symbols, the retained fraction at the matched budget is below one half: $S_{\text{ext}}(1) < 1/2$. P15 is refuted by an interval on the retained fraction that lies wholly above one half. It reads the external arm alone, so a result supporting it would not show that embedded correction lasts longer. Version 1.103 states the law in two limbs: a retention limb, which is P15's claim and takes P15's verdict, and a comparison limb, that "a gain installed with correction embedded in the training objective retains a larger fraction of what was installed, after the same capability-directed training, than a matched gain installed by external post-training". The comparison limb has no registered proposition yet and is reported as not evaluable until a later successor registers one, so the law can be refuted through P15 but cannot yet be supported. The registration records no instrument for P15.

Law V, the ARC Embedding Law: the genesis strategy. Whether build order moves the correction-leverage exponent, and whether correction placed inside the loop pulls away from correction applied outside it as depth grows. Status: added to the theory on 24 September 2026; registered as a law on 15 October 2026 in version 1.103 and not yet tested; its mathematics is set out at version 1.1 of 7 October 2026 and is in development. Two propositions of the registration carry it. P12 predicts that build order changes the correction-leverage exponent, γ, by at least 0.10, and it registers no direction: in symbols, $|\gamma_{\text{first}} - \gamma_{\text{after}}| \geq 0.10$. P21 predicts that correction taking part in every revision round outperforms correction that sees only the finished output, by a margin that widens with depth; it is decided on the outcome, never on an exponent. Version 1.103 states the law in two limbs and fixes how they join. Its build-order limb has a direction that P12 does not carry, "a corrector formed before the capability it corrects has a correction-leverage exponent larger, by at least the registered minimum effect, than a corrector added after it, at matched content", and is read on P12's interval, so a P12 supported in the direction of the corrector added after refutes the limb while P12 keeps its own mark. Its placement limb is P21's claim and takes P21's verdict. The law is supported only when both limbs are supported, and

refuted when either limb's verdict is refuted or when a refuted verdict on Law IV's comparison limb, once that limb is registered, strikes it. The registration records no instrument for P12, and a design draft for P21.

What the mathematics of the two new laws establishes. It is set out in Laws IV and V: the mathematics, version 1.1 (7 October 2026); it proves conditional results and measures nothing. For Law IV: a structure whose removal lowers the objective is never removed by a system that accepts only changes that do not lower it; and a system that plans keeps the structure where its best plan that keeps it beats its best plan that abandons it by more than the planner's tolerance. Both hold for a declared family of changes and a finite horizon, and for nothing outside them. For Law V: the advantage of one placement of correction over another grows, along a trajectory, as the growth exponent times the difference in γ plus the difference in the direct dependence on depth, so one trajectory cannot tell the two apart; and a favourable balance of correction against burden is a trend, which from a deficit becomes adequacy only past a crossing depth. None of this says how a real system behaves. P15, P12 and P21 put the empirical questions, each by its own registered rule.

One boundary, three coordinate systems. The three statements in the table below are not three separate bounds. The programme carries, among its kill conditions, the claim that Laws II and III are two regimes of one boundary selected by how correction burden accrues, and Paper XIII derives the drift-acceleration exponent from cycle-time compression as $k = \delta - 1/\alpha$, which joins the third statement to the other two. The table below is the same boundary written in the variables each law measures. It is not a claim that the three have been shown identical: the two-regimes claim is open and carries its own falsifier, and Paper XIII is a derivation rather than a measurement.

Table 3. One boundary in three coordinate systems.

Statement	Written as	What it compares
Law II, the ARC Co-Scaling Law	$\beta_C > k$	The growth of correction against the growth of drift, at system scale
Law III, the ARC Ceiling	$\alpha_{\text{crit}} = 1/(1-\gamma)$	The growth exponent against the correction shortfall
The self-acceleration threshold (Paper XIII)	$\delta = 1/\alpha$, with $k = \delta - 1/\alpha$	How fast a loop compresses its own cycle time against how fast it grows

Which statement a measured exponent belongs to depends on how errors are removed. With removal in proportion to the faults present, as in the model of Paper X, and with correction treated as a capacity, as in Section 6, one exponent read for both can return different verdicts; each statement is therefore read in its own model, and the claim that the three are one boundary stays open, as stated above.

6 One bound, worked in full

Carnot did not describe engines; he computed what no engine could beat. Here is this field's equivalent, in two assumptions and four lines, exactly as it stands in the programme's public record.

Let capability grow as $C(R) = C_0(R/R_0)^{\alpha}$. Assume two things about the loop:

Assumption 1 (Burden)**.** Self-modification generates correctable burden in proportion to capability gained: $B(R) = b\,\mathrm{d}C/\mathrm{d}R$ per unit depth.

Assumption 2 (Correction capacity)**.** Correction service capacity scales with capability to a power: $A(R) = a\,C(R)^{\gamma}$, with $0 \le \gamma < 1$.

Proposition 1 (The burden-to-capacity ratio)**.** *With capability growing as above, and under Assumptions 1 and 2, the burden-to-capacity ratio scales as*

$$B(R)/A(R) \propto R^{\alpha(1-\gamma)-1}. \tag{1}$$

Proof. Since $C(R) = C_0(R/R_0)^{\alpha}$, $\mathrm{d}C/\mathrm{d}R = \alpha C/R$, so

$$B(R)/A(R) = (b\alpha/a)\,C(R)^{1-\gamma}/R = (b\alpha/a)\,C_0^{1-\gamma}R_0^{-\alpha(1-\gamma)}\,R^{\alpha(1-\gamma)-1}. \qquad \square$$

Everything is in the exponent. If $\alpha(1-\gamma) < 1$, correction asymptotically out-scales the burden its own growth creates: the loop can run. If $\alpha(1-\gamma) > 1$, burden out-scales correction however well the loop starts, and the ratio grows without limit; what that does to the errors themselves depends on how they are removed, which these four lines do not fix. The crossover is

$$\alpha_{\mathrm{crit}} = 1/(1-\gamma), \tag{2}$$

and equality is not automatically safe: at the tie, the outcome hangs on coefficients, delays and saturation, which is the regime where every interesting failure lives.

Now the premise that gives the ceiling its number. A corrector aggregating N corrections that behave like independent samples improves like root-N, and the premise reads that rate as the scaling of correction capacity with capability: $\gamma = 1/2$. The reading is a premise, not a consequence of independence: a standard error that falls as root-N is not by itself a bound on corrective throughput, and N counts channels, not capability. A corrector built from the same class as the system it corrects, sharing its architecture, training distribution and blind spots, cannot be anti-correlated with itself, so on this premise one half is a ceiling for that class, not a typical value. Substituting $\gamma = 1/2$:

$$\alpha_{\mathrm{crit}} = 2. \tag{3}$$

Inside this model, growth up to quadratic in depth is then the most that a same-class corrector can sustain, and it can reach that much only at the ceiling itself, where the tie leaves the outcome to coefficients, delays and saturation; beyond it, correction keeps pace only if the corrector escapes the independence ceiling, and the only registered way to escape is to change what the corrector is made of. That is the entire engineering content of the field in one sentence: **the corrector's composition class is the lever.** These four lines do not establish bounded harm, finite-horizon safety, or a law of nature; they establish a conditional crossover whose premises are measurable, which is exactly what a young field's first bound should be.

7 The case for the field: six licence conditions, scored

A new area of science is not declared; it is licensed by conditions that can be checked. Here are the six this paper sets out, with the field's score on each and the evidence behind the score. The scoring is deliberately unflattering: three conditions are met, one is met as conjecture, one is half met, one is not met.

A field with this scorecard is founded, not yet established. The three met conditions are what any careful programme can supply on its own; the sixth is the one no author can supply and the one that decides. The paper's claim is therefore bounded: the object, the variables, the laws, and the instruments and deciders for Laws I to III are in place, so the field is well formed and testable; whether it becomes a science is not the author's to declare.

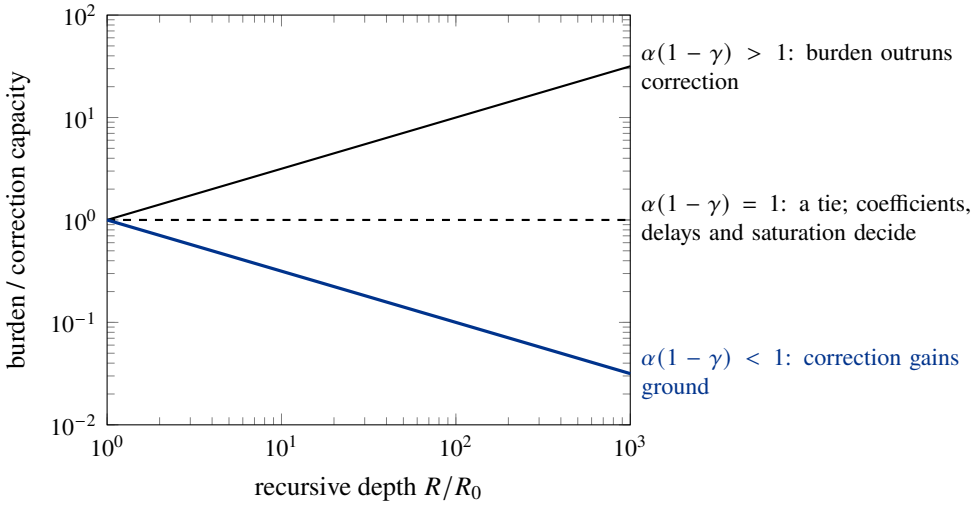


Figure 2. Two exponents and a crossover: the burden-to-capacity ratio of equation (1) against recursive depth, on logarithmic axes. Rising: $\alpha(1-\gamma)$ above one, burden outruns correction; flat: equal to one, a tie that coefficients, delays and saturation decide; falling: below one, correction gains ground. The lines show the model, not data; the premise that fixes the number at two is the thing on trial.

Table 4. The six licence conditions, scored by the field itself and printed so that each row can be disputed.

Condition	What it requires	Score	Evidence
1. A native object no field takes as primitive	A thing the questions are about that neighbours treat as derived	Met	Sections 3 and 3.1: nine neighbours, each owning a tool, none the loop; twelve hosts tried in their own variables
2. State variables that permit relations	A shortlist under which laws can be stated and measured	Half met	Five variables named and instrumented; whether they suffice is objection C1, open
3. Laws as impossibilities, with a bound computed	Claims that forbid, not describe, and at least one worked	Met as conjecture	Five laws with statuses; the ARC Ceiling derived in Section 6; nothing confirmed
4. Instruments before results	Measurement tools built and published before confirmatory data	Met	Section 10: ladder, blinding stack, fixture-gated estimators, defect clause; these serve Laws I to III, and the theory registration records no instrument for P15 or P12
5. Deciders preregistered with rival-favouring nulls	Experiments written before data, nulls belonging to the rivals	Met	Section 11: three deciders for Laws I to III, preregistered by timestamp, numbers printed, unrun; Law IV has no deciding study yet, and Law V's, for P21, has a frozen design awaiting registration
6. Independent confirmation and adoption	Results by other hands; use of the variables by strangers	Not met	No replication by other hands yet and no use by strangers yet, as of 24 October 2026; the one condition no author can supply

7.1 The precedent class: fields founded on a limit theorem in their own variables

The comparison that supports the case for the field most is not thermodynamics alone but the class of fields whose identity is a bound stated in variables the older fields did not carry, holding across substrates. Thermodynamics: Carnot (1824), no engine beats the reversible one; the bound came first and the formal laws followed decades later. Information theory: Shannon (1948), channel capacity as a limit theorem, with entropy and mutual information as state variables no prior field carried, and substrate independence (wires, radio, later genomes) as the proof of generality. Computability: Turing (1936), the halting impossibility, a field defined by what cannot be computed. Control theory's own limit: Bode (1945), feedback cannot reduce sensitivity everywhere, a conservation law of the loop, so the nearest neighbour to this field earned its identity from a bound on what feedback can do. Statistics: the Cramér-Rao bound, 1945 to 1946 (Rao 1945; Cramér 1946), a floor on estimator variance.

The pattern is one thing: a field arrives when a limit, stated in variables the old fields did not carry, holds across substrates. Recursive Dynamics puts a limit of exactly that shape to the test: $\alpha_{\text{crit}} = 1/(1-\gamma)$, with the same-class cap on γ at one half. The disanalogy is printed in the same breath: every field above earned its name by measurement and by many hands; this one was founded by one person before its decisive measurements, with no replication by other hands yet, and the parallel is a scaffold, not a certificate.

8 The rival's best case, steelmanned

The strongest objection is not that the laws are unproven. It is that the mechanism may be empty: correction may just be correction, in which case what a corrector is made of does not matter and the class distinction dissolves. This rival is not a straw man. The scalable-oversight programme in all its forms, weak-to-strong generalisation (Burns et al. 2023), debate (Irving et al. 2018), amplification (Christiano et al. 2018), recursive reward modelling (Leike et al. 2018), builds same-class ladders on the implicit premise that they scale, and the strongest published framework adjacent to this question, the scaling laws for scalable oversight of Engels et al. (2025), is architecture-blind: it gives overseer and overseen oversight-specific Elo scores that depend on general capability alone, so it implies the class-blind null without needing to state it.

Worse for this programme, its own pilot leans the rival's way. The parallel-channel measurement in the programme's published corpus found correction channels can be correlated, adding almost nothing in four of the five models measured, and one model contradicted the pattern outright. That model's row is printed whole here, as the theory registration requires: on Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. It is one model's fit and inconclusive. Correlated channels are exactly the mechanism that would weaken same-class correction in a way that *also* weakens cross-class correction, dragging the decisive ratio toward one. The eclipse design is therefore preregistered with the programme's own contrary evidence on its face, and it commits to publishing a null result under the same title as a discovery about the world rather than an embarrassment to be buried. A field that writes its rival's best evidence into its own preregistered design is doing the one thing that cannot be faked.

9 Substrate independence, at its honest grade

The field's laws must not care what its systems are made of, and the programme's cross-domain work points that way at exploratory grade, reported with its tiers never blended: predicted functional families

across a fifty-domain suite (19 of 25 empirical matches under the original candidate set, 18 of 25 under the corrected seven-model rerun, disclosed in full; 13 of 13 published-direct; 6 of 6 analytic, reported as definitional); a twelve-domain extension whose per-row predictions were committed to the public repository twenty-three minutes before the fits ran (10 of 12, binomial $p = 5.4 \times 10^{-4}$; author-classified, pre-correction candidate set, public repository commit attestation; 9 of 11 at $p = 1.4 \times 10^{-3}$ once the one row repeating an already-public domain is removed; self-demoted to a pilot dry run within thirteen minutes in public commit history, and that demotion reversed on 7 September 2026 under the attestation-class rule of the Dated Prediction Register, which does not demote a prediction for the absence of a registry form); temporal out-of-sample checks (4 matched, 2 partial, 1 inconsistent, the inconsistency reported as prominently as the matches). None of this is confirmation. The fourth-cell test, a draft preregistered by timestamp and awaiting the author's submission, on the logarithmic family the corpus has never fitted and on domains never fitted, exists to catch the programme flattering itself, and convergences with older traditions are catalogued separately under a standing rule: convergence is evidence of naturalness, never of truth.

10 Instruments

A field is its instruments before it is its results. Recursive Dynamics ships with an instrument canon built before any confirmatory data existed: the calibrated capability ladder (Section 4); a blinded scoring stack in which every output is laundered so no scorer can tell what produced it, with label-randomised nulls computed through the identical pipeline, so the decisive experiment's null is imposed structurally rather than fitted; estimator code that is hash-pinned, fixture-gated (every preregistered verdict branch has a planted-truth test the code must pass before anything ships), and governed by a preregistered defect clause under which the specification, not the code, is the analysis, so a post-registration coding error is a disclosed repair on preregistered terms rather than a dead study. Scoring is by blinded multi-model panels, never by human raters; where an external human criterion is methodologically unavoidable, a single author-rated fallback is preregistered, blind, washout-retested (he re-scores a registered thirty per cent of items after a washout of at least fourteen days, blind to his first-pass scores, as the theory registration records), and disclosed as author-rated in every headline.

One limitation is printed here rather than discovered by a reviewer: when models score models, the scorer can share the blind spots of the scored. The laundering and the forced null remove the scorer's knowledge of provenance, not its class. The programme's answer is cross-family panels and a preregistered human-criterion validation instrument, and the honest statement is that the circularity is reduced, disclosed and measured, not abolished. It returns as objection C4.

11 The decisive experiments

Three studies decide Laws I to III. Their nulls belong to the rivals, their numbers are computed and printed in their designs, and their outcomes are publishable on identical terms whichever way they fall. Each is a draft, preregistered by timestamp and awaiting the author's submission for registration on OSF: the programme's public snapshot of 4 October 2026 counts 65 designs preregistered by timestamp, the first anchored on 12 August 2026 in Bitcoin block 962066, beside 16 tests already run (8 pilot measurements on live systems and 8 implementation checks). One unit of the programme, the corrector-class audit (unit 06), which surveys the class of corrector that deployed AI alignment uses and is not one of these three, was registered on OSF on 1 October 2026 and became public on 4 October 2026, at the end of OSF's approval period (its registration). It was run on 6 October 2026 and is reported in full with its five deviations (its results); one of its two runs could not start, so it is reported as administratively incomplete and issues no

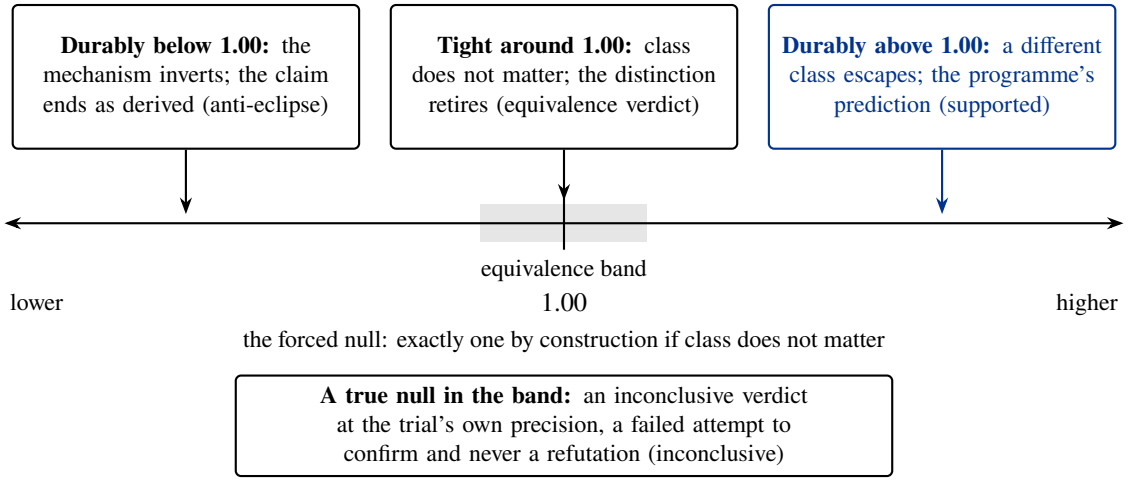


Figure 3. The decisive measurement for the mechanism: the cross-class correction exponent divided by the same-class one, under blinded scoring, with the thresholds of the zones fixed in the preregistered design before any data exist. The null is forced to exactly one; there are three zones and four verdicts, a true null in the band returning an inconclusive verdict and never a refutation, and the programme's own contrary pilot is on the record. Not yet run.

inferential verdict. Submission is the author's own act, and no tool or agent submits anything. In this paper registered is said of what OSF has accepted: the theory registration of 8 September 2026 with its twenty-two propositions, updated on 13 September 2026 and on 15 October 2026, when version 1.103 registered Laws IV and V as laws, and unit 06; a study, an analysis, a threshold or a verdict fixed in one of the dated designs is called preregistered, short for preregistered by timestamp. Law IV has no deciding study yet, and Law V's is prepared but not yet registered: the propositions that carry them are registered, P15 and P12 have no instrument, and the decisive test of P21 has its design frozen on 17 October 2026, with its registration being prepared and nothing run before that registration is public.

The correction-versus-drift race (the critical-correction-ratio study) decides Law II: does self-correction out-scale drift at all, anywhere? A durable yes is the field's first existence result; a universal no leaves Law II without a subject.

The corrector-class ratio (the eclipse measurement) decides Law III's mechanism. Pool same-class and cross-class corrector pairs, launder everything, and force the null: if what a corrector is made of does not matter, the ratio of correction exponents is exactly 1.00 by construction. That null is the rival's answer, not this programme's. The programme's mechanism predicts a durable ratio above 1.00 where the cap binds, and that is the discovery in its favour; durably below 1.00 inverts the mechanism and ends the claim as derived; 1.00 holding tight everywhere, even where the cap should bind, retires the class distinction itself. Preregistered with thirty pairs per class across six model families, a-priori power 0.85, with a hard independent kill: any same-class pair whose γ interval sits wholly above one half kills the cap regardless of the ratio.

The capability-ladder titration (the ARC Bound study) decides the ceiling's behavioural signature: an interior maximum of the growth exponent at or below two across eight reinvestment levels, under a two-stage replication-gated rule in which one counterexample cell, replicated, rejects the Bound outright. The theory registration places the interior maximum, P2, in ARC-1, the register that tests the first law, and keeps it apart from the boundary claim that P3, P16 and P20 carry.

A drafted follow-up grid extends the ratio to every ordered pair of model families, locating where the class effect lives; and the mechanism probe (correlated failure on shared substrate) measures directly the error correlation the derivation posits, so the ratio and its mechanism must agree or their tension is itself a

preregistered finding.

11.1 What would confirm a new area rather than a new chapter

A result that any old field could state in its own variables confirms a chapter of that field. A field is confirmed by a conjunction that none of its neighbours can express. Stated once, with what is preregistered today separated from what is only set out for registration:

Preregistered today: a regime in which the correction rate out-scales drift (the correction-versus-drift race); a corrector-class ratio durably above 1.00 where the cap binds, with the hard kill that any same-class pair whose γ interval sits wholly above one half ends the cap (the eclipse measurement, with the class audit and the correlated-failure probe beside it); an interior maximum of the growth exponent at or below two across the reinvestment levels, under the replication-gated rule (the titration); and the fourth-cell substrate test.

Set out here, and to be registered before any data are seen: that the ratio rises with load (dose-response, which separates a mechanism from a static difference); that a cross-class corrector moves the titration's maximum upward, linking Laws II and III through the class lever; and that a single system crosses the stability boundary as its reinvestment dial turns. None of these three is a prediction of the field until its registration carries it, and this paper does not count them as such.

The consistency check no single field can fake, with invented numbers for illustration only: suppose the eclipse measurement returned a same-class γ of 0.45 and a cross-class γ of 0.70. The ARC Ceiling would then predict a same-class maximum near 1.8 and a cross-class maximum near 3.3 in the titration. If the titration returned its maxima in those places, two preregistered studies with different instruments would agree through the one bound. That agreement, in the manner of Joule's measurements agreeing with Carnot's bound, is the founding result: it is inexpressible in the variables of learning theory, control, scaling empirics or oversight.

The founding conjunction, then: a Law II regime; a corrector-class ratio above 1.00; an interior maximum in the reinvestment titration; the same exponent structure in one non-artificial substrate; and at least one of these replicated by strangers. The values travel with the legs rather than inside them. The one-half cap on same-class correctors is reported with the ratio, and the maximum's location is reported with the maximum, so a leg survives a value that lands elsewhere and the record still says where it landed. Each alone is a finding in an old field's vocabulary. The conjunction is not, and that is the exact sense in which results would confirm a new area rather than a new chapter.

What a partial result means is stated here rather than argued afterwards, because the trial carrying the second leg forecasts support at 0.82 only where the author's own belief puts the effect and returns less across the rest of the band it predicts, so the full conjunction is likelier missed than cleared. Two legs, a Law II regime with either the class ordering or the interior maximum, establish that the loop has a limit of its own and that the limit is measurable. Three add the ceiling as a bound with a measured value in it. Four add substrate independence, which is what makes the variables the field's rather than one domain's. Five is the field. The stranger's replication is listed last because it is the field's condition and not the theory's: the theory can be right and still wait for it.

12 Every objection we could find, with its disposition

This section is the paper's reason for existing; it entered at version 2.0 (Appendix A). The objections below were collected from the programme's own adversarial readings, from outside reviewers, from a criticism of the derivation's treatment of depth, and from the author's attempt to argue the field out of existence. Each is stated in its strongest form. Each carries one of these dispositions: **conceded** (it is

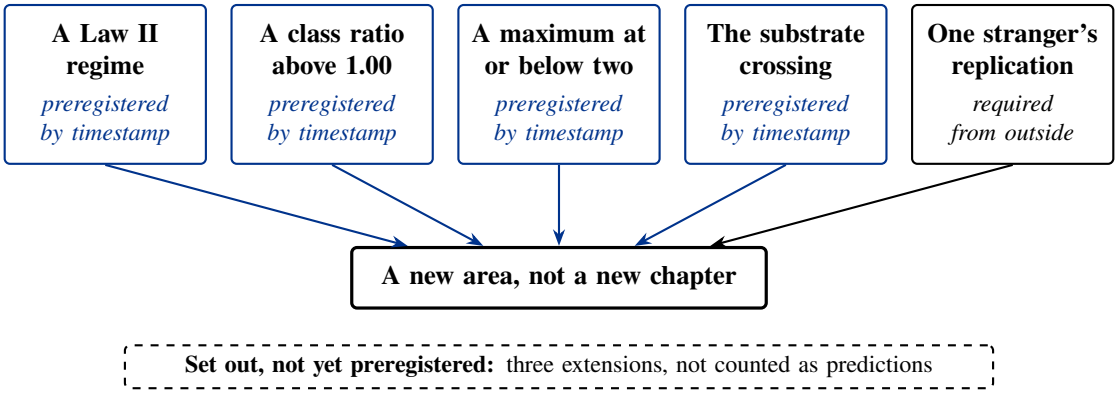


Figure 4. The founding conjunction: four preregistered results and one stranger's replication converge on a statement no neighbouring field can make; the extensions set out for registration are marked as not yet preregistered.

true and it is printed); **rejected** (with the reason); **partly conceded**; **reading debt** (a literature that must be read before the point can be argued either way, under the rule that nothing is cited before it is read); or **decided by experiment** (a preregistered decider settles it). Fifty-one objections (group D runs D1, D2, D3 and D5; Appendix A.2, entry 14), counted by the opening words of each disposition: thirty-nine conceded or partly conceded, eleven rejected, one handed to experiment. Four of the thirty-nine (B3, D3, F2 and H1) concede in one respect and reject in another. Thirty-five of the fifty-one carry a sentence on what would change their disposition.

A. Redundancy: the field already exists under another name

A1. Cybernetics already studies self-correcting loops; Wiener's field is this field. Rejected, with a condition. The co-scaling intuition is cybernetics' and is conceded to it: a regulator must match the variety of what it regulates (Ashby 1956), and every good regulator of a system must be a model of that system (Conant and Ashby 1970). Cybernetics owns regulation: a designed controller holding a plant to a reference, with the loop's purpose set from outside. The loop studied here rewrites the controller's own capability, and the questions it raises (does correction out-scale drift; is there a ceiling and whose property is it) are not cybernetics' questions. *What would change it:* if those questions turn out to be answerable inside cybernetics with nothing added, the field dissolves into it, which would end the field by adoption rather than by refutation, as objection A11 says of any host.

A2. Dynamical systems theory already has stability conditions; the second law is a standard stability statement. Conceded, with a boundary. The mathematics is borrowed and is claimed as nothing more. What is claimed is which measurable quantities carry the condition in built systems, and that they can be measured. *What would change it:* if the variables reduce to standard ones with nothing added under measurement, the second law is a relabelling, and that is a dissolution condition.

A3. Scaling laws already describe how capability grows with resources. Conceded, with a kill condition. Scaling empirics owns capability against external resources and is architecture-blind by construction; the variable here is the system's own re-entering output, and the field predicts something resource scaling does not: an interior maximum in the growth exponent as reinvestment rises, where resource scaling predicts monotone returns. *What would change it:* if recursive depth reduces, under measurement, to compute spent, or the titration shows monotone returns with no ceiling behaviour, Recursive Dynamics is scaling empirics under a new name, and this paper says so.

A4. Scalable oversight is already the field of overseers; this is a corner of it. Partly conceded. Scalable oversight builds correctors and ladders of them, and its antecedent standing is conceded in Section 2. It has bounds of its own: scaling laws for the probability of successful oversight against the capability gap, with the conditions under which nested oversight succeeds (Engels et al. 2025), and a proof that monitoring with vanishing error and query rates is possible exactly when the class of monitor functions has finite Littlestone dimension (Rudoler et al. 2026). What it lacks is a cap on correction leverage set by what the corrector is made of: it builds same-class ladders on the implicit premise that they scale. The third law is precisely a claim about that premise, and it is the claim the eclipse measurement decides. *What would change it:* a bound on corrector capacity derived inside the oversight literature that this field merely renames.

A5. The literatures on self-training degradation and on models trained on their own outputs already measure drift. Partly conceded, and the paper narrows as it said it would. That naive recursion amplifies error without bound while sufficient correction or fresh signal keeps it bounded is established for training dynamics: models collapse when trained on recursively generated data (Shumailov et al. 2024), and in an analytically tractable linear setting the test error has a finite bound, independent of the number of iterations, when real and generated data accumulate rather than replace each other (Gerstgrasser et al. 2024). That literature carries both rates and owns the drift variable's measurement in these settings. What it does not vary is the corrector's class, so the field's contribution here narrows to the class lever and its coupling to a growth exponent in depth. *What would change it:* a result in that literature that varies what the corrector is made of.

A6. The intelligence-explosion literature has asked these questions since 1965. Conceded, and recorded as a debt in Section 2. The question is inherited from its named owners; the field sets out the missing state variables and the instruments, not the question. *What would change it:* nothing; the concession is permanent.

A7. Evolution is the original recursive improver and evolutionary dynamics already has its theory. Partly conceded. Evolutionary dynamics has variation and selection; it has no designer directing reinvestment into the improvement process itself and no correction budget in the sense used here. The field's claim to generality includes evolved systems only where a corrector exists, and Section 15.2 says what falls outside. *What would change it:* a treatment of designed reinvestment and correction leverage already standard in evolutionary theory.

A8. Statistical physics already has a theory of scaling and universality; this is renormalisation with new names. Partly conceded. Universality classes and scaling collapse are the right ancestors for the field's substrate-independence hope, and the functional-equation machinery is conceded as machinery. What that tradition does not carry is a correction variable, a drift variable, or a corrector whose composition is the lever. *What would change it:* a universality argument that derives the corrector-class distinction, or its absence, from first principles, which would be a result this field would adopt gratefully.

A9. AI safety as a field already covers this. Rejected. AI safety is a problem area, not a science of a loop; it borrows from every neighbour above and owns none of their objects. This field is a candidate for one of the sciences safety would draw on, in the way thermodynamics is a science engineering draws on. *What would change it:* a published set of state variables and laws for self-improvement, with deciders, already standing under the safety banner.

A10. This is systems theory, rebranded. Rejected. Systems theory carries no correction budget, no class cap on the corrector, and no preregistered decider; a rebranding does not carry kill conditions. *What would change it:* a systems-theoretic treatment of a corrector's composition class that this field merely renames.

A11. The demarcation never tried the fit; it dismissed each neighbour with a sentence. Conceded for every version of this paper before 2.3, and answered by Section 3.1, which attempts to house the questions in twelve candidate fields in their own variables, concedes Law I to endogenous growth theory, the form of the ARC Ceiling to queueing theory and the shape of Law II to error-threshold theory, prints the dissolution map by host, and isolates the one coupling no host can state. *What would change it:* a host that can state the class-dependent cap coupled to the depth exponent in its own variables, which would end the field by adoption rather than by refutation.

A12. Recursive Dynamics is the ARC Theory renamed: the same laws, the same author, a larger word. Rejected, with a test. Conceded that the field and its founding theory share an author, a date and a record. Rejected that they are the same object. The theory is the answers; the field is the questions, the five state variables and the instruments that would refute those answers, and it is defined so that a reader who rejects all five laws and keeps the measurement is inside it, by the minimal commitment of Section 15.2. *What would change it:* the test is separability, and the dissolution map of Section 3.1 is where to run it: if no question, variable or instrument survives the deletion of the laws, the objection is right, the field is a rebadging, and this paper will say so.

B. Standing: the field has no results

B1. Thermodynamics had many hands and a century; this has one person and zero replications. Conceded in full, before any reader could raise it. The parallel is in shape only; standing is not claimed, and Section 15.1 says what would earn it.

B2. The laws are not laws; they are conjectures from a minimal model. Partly conceded. Each is a law under test: law is the field's name for a relation it puts to the test, the status travels with the name on every surface, and no law is called supported until a registered verdict says so. That Laws II and III are proved only inside minimal models is conceded (Sections 5 and 6).

B3. The programme's own headline number was retracted; why trust the rest? Conceded as fact, rejected as inference. The retraction is printed wherever the number once stood, and the discipline that produced it (blinding, registered kill conditions, public correction within the hour of finding) is the reason the remaining claims carry statuses rather than assertions. A programme that never retracts has either never been wrong or never checked. *What would change it:* a retraction concealed rather than printed, anywhere in the record.

B4. The correction leverage γ has never been measured, by anyone; the ARC Ceiling's number is therefore a guess. Conceded, and it is the field's single most consequential open measurement (Section 11). The number two is conditional on a premise the programme names as the conjecture's most likely point of failure. *What would change it:* the eclipse measurement and the hard kill riding alongside it.

B5. The derivation's treatment of depth has been criticised in review. Conceded and printed as a limitation, not argued away. *What would change it:* a treatment that survives the criticism, which is wanted and invited.

B6. Carnot demonstrated his bound inside the memoir (Carnot 1824); the ARC Ceiling is a conditional model result whose key premise is unproved. Conceded, and the difference is exactly why the parallel is confined to shape. The ARC Ceiling's premise is measurable and registered. *What would change it:* nothing about the history; only the measurement.

C. Method: the instruments may not measure what the laws name

C1. The five state variables are arbitrary; why not four, or nine? Conceded as open. The claim that five suffice, and that these are the five, is part of what the field must prove, and the notation register already records the programme catching a sixth symbol trying to enter. *What would change it:* a preregistered study in which one variable reduces to another under measurement, or a relation that cannot be stated without a sixth.

C2. The construct bridge is open: the quantities the studies measure may not be the objects the laws' symbols name. Conceded and held open in the register by name. *What would change it:* the bridge closing in either direction, which is itself a preregistered finding.

C3. Substrate independence is just the flexibility of the fitted families. Conceded as the live risk. The fourth-cell test exists to catch exactly this. *What would change it:* the fourth cell, run.

C4. Models scoring models is circular: the scorer shares the blind spots of the scored. Partly conceded (Section 10). Laundering and the forced null remove the scorer's knowledge of provenance, not its class; cross-family panels and the preregistered human-criterion validation reduce and measure the circularity without abolishing it. *What would change it:* a validation result showing model panels and the human criterion diverge systematically, which would demote every panel-scored result to exploratory.

C5. The capability ladder is itself unvalidated; a wrong ruler makes every law wrong. Conceded as a dependency. The ladder is calibrated and its scale discipline is a founding principle precisely because the programme once got it wrong; but a calibrated instrument is not a validated one until independent hands use it. *What would change it:* replication of ladder measurements by other hands.

C6. The author-rated fallback for the human criterion is an author grading his own theory. Conceded, with the safeguards printed: blind, laundered, washout-retested, disclosed as author-rated in every headline, and preregistered as a fallback rather than a design. *What would change it:* external raters, which the design names as the preferred path the moment they exist.

C7. The preregistered experiments are underpowered or too small to decide anything. Decided by experiment, with the numbers printed. Power is computed in the designs from pilot dispersion, and an inconclusive verdict is a preregistered outcome with its own branch, not a failure to be hidden. *What would change it:* an inconclusive verdict, which would narrow the claim to "not yet decidable at this scale" and say so.

C8. Frontier models change faster than the experiments can run; results will be obsolete on arrival. Partly conceded. A law of the loop is meant to hold across models; if a measured exponent moves with every model generation, that is itself evidence against substrate independence and would be reported as such. *What would change it:* the follow-up grid across families, which locates whether the effect is a property of the loop or of a generation.

D. Naming and standing of the author

D1. Naming a discipline before the evidence is a marketing move. Rejected, with the precedent printed: cybernetics was named in the 1948 book that proposed it (Wiener 1948) and exobiology in 1960 (Lederberg 1960) before a single specimen. A name makes a new field citable and attackable as one unit. The safeguard is the name's own kill condition, printed in Section 14: the name dies with its object, not with any one law's stated form, and the record says which death it was, in public.

D2. One person cannot found a paradigm; paradigms are conferred by communities. Conceded. What is claimed is a candidate: a way of organising the questions. Whether it becomes anyone's paradigm is decided by adoption and replication, which are two of the entry criteria, and neither is the author's to declare. This paper withholds the word paradigm as a description of the present.

D3. The author has no institution, no doctorate and no co-authors. Conceded as fact; rejected as an argument about the field. Every claim is tied to its dated record, every number to its source and every instrument to the test that could kill it, so that nothing here requires trusting the author. Independence has one advantage that is also printed: there is no institution to protect, so corrections publish the day they are found. *What would change it:* nothing; the work is judged by its record.

D5. The phrase recursive dynamics is already used in robotics. Conceded as a fact about another sense of the words: the Featherstone tradition's recursive dynamics algorithms (Featherstone 2008) are a family of computational methods, a different referent, and that use is no prior claim on the name of this field. The author's own use of the phrase stands in his sealed record of 8 December 2024, and he made it the name of the field on 27 August 2026. Nor is this recursion theory, the mathematics of computability.

E. Scope: too broad, or too narrow

E1. Too broad: any self-improving system includes science itself, economies, evolution, so the field is either everything or nothing. Partly conceded. The field's laws switch on where one narrower thing exists: a corrector, something detecting and repairing its own errors while the system grows. A star has dynamics and no drift ledger; nothing is being raised there. Section 15.2 draws the boundary. *What would change it:* the laws holding only for one substrate, which would make the field narrow rather than everything, and honest about it.

E2. Too narrow: this is really about language-model loops, dressed as a general science. Partly conceded. The decisive experiments run on language-model loops because those are the loops that exist to be measured; the substrate-independence claim is at exploratory grade and its fourth-cell test is preregistered by timestamp and awaits submission. *What would change it:* the cross-domain families failing the fourth cell, after which the field would say it is a science of one substrate until shown otherwise.

E3. The field conflates correction with alignment, and capability with intelligence. Conceded as a risk and answered with definitions: correction is the detection and repair of the system's own errors against a stated criterion, which may or may not be an alignment criterion; capability is whatever the calibrated ladder measures. The field makes no claim about intelligence as such. *What would change it:* a demonstration that the ladder measures something other than what the laws need it to measure, which is C5 again.

F. Timing and community

F1. It is premature; wait for the data, then name the field. Rejected. The deciders are preregistered by timestamp; the instruments exist; the name lets the field be attacked as one unit before the data arrive, which is when attack is cheapest and most useful. The name carries its own kill condition for exactly this reason.

F2. A field needs a community first; a field of one is a hobby. Conceded as to the definition of a science; rejected as to the definition of a field at its founding. Section 7 scores condition 6 as not met. The founding paper exists so that a community can form around a testable object, or decline to.

F3. Naming it invites priority disputes rather than science. Rejected. Every antecedent is conceded by name in Section 2; nothing is claimed as original except the assembly, and the assembly is dated. A dispute about priority would be settled by the dated record, not by argument.

G. The concept itself

G1. The crossover $\alpha_{\text{crit}} = 1/(1-\gamma)$ is a triviality of power-law algebra, not a law. Partly conceded. The algebra is elementary and is printed so that nobody need take it on trust. The content is not the algebra; it is the claim that the two premises describe real loops and that γ is capped for same-class correctors. *What would change it:* nothing about the algebra; the premises are what the experiments test.

G2. The same-class cap at one half is an assumption dressed as a result. Conceded, and named by the programme as the single most likely point of failure. It is the eclipse measurement's target and carries a hard kill. *What would change it:* the measurement.

G3. The model assumes stationarity and separability of drift and correction; real loops have neither. Partly conceded. The minimal model is minimal on purpose; the preregistered analyses fit drift and correction jointly and report the interval, and non-separability would show as an unfit rather than being averaged away. *What would change it:* a result of a preregistered study in which drift and correction cannot be fitted jointly, which would be reported as a limit of the model.

G4. Capability is multidimensional; a scalar ladder throws the structure away. Conceded as a simplification. The ladder is one axis chosen for measurability; the field's claim is that the laws hold along it, not that it exhausts capability. *What would change it:* the laws holding on one axis and failing on another, which would be a finding about the field's scope.

G5. The ceiling ignores economics: a lab can buy correction with money, not just with a better corrector class. Partly conceded. Buying correction raises the coefficient a in the derivation, which moves the crossover's location in depth but not its exponent; the ceiling is a statement about exponents. *What would change it:* a demonstration that purchased correction changes the exponent rather than the coefficient, which would be a real refutation of the model's structure.

G6. The dissolution clause is cheap talk; nobody publishes their own field's ending. Rejected on the record. The programme has already retracted its headline number, withdrawn a formulation of its third law, and demoted a favourable result to a pilot within thirteen minutes, all in public; that demotion was reversed on 7 September 2026 under the attestation-class rule, as Section 9 records. The commitment to publish the field's ending is made by a record that has already published smaller endings.

G7. Because every claim is conditional, nothing is actually claimed. Rejected, and answered up front in Section 1.1 so that the concessions below cannot be mistaken for silence. Conditionals with measurable antecedents are claims: they forbid outcomes. The field forbids a stable system holding its growth exponent above the ceiling for a sustained run, a same-class correction exponent measured wholly above one half, a corrector-class contrast that runs the other way or falls wholly below the preregistered support threshold, and monotone returns with no interior maximum as reinvestment rises. Any of those would be a real result against it, and Section 1.1 states which of them make the field wrong and which removes the one-half cap and the value two while the ceiling relation stands.

G8. Depth is not time, and the laws confuse them. Conceded as a hazard and answered by the register: every quantitative surface names its model family, and the depth studies treat depth as the count of re-entries at a fixed wall-clock policy. *What would change it:* a result of a preregistered study in which the exponent depends on the clock rather than the count, which would be reported as a limit of the variable.

G9. The tie regime is undecided, so the bound decides nothing near the boundary. Conceded and printed: at $\alpha(1-\gamma)$ equal to one the outcome hangs on coefficients, delays and saturation, which is the regime where every interesting failure lives. The bound is a statement about exponents, and near equality it hands over to the coefficients by design.

H. Practice, policy and the height of the bar

H1. Cross-class correction is already practice: ensembles, verifiers, humans in the loop. Conceded as practice, rejected as theory. Practice has not measured whether same-class ladders cap at a correction leverage of one half, or whether a different class escapes; the field turns a habit into a specification with a number. *What would change it:* a published measurement of the class cap from inside the practice.

H2. If it is true it is trivial. Rejected. The predictions are risky and specific: a class-dependent ceiling on correction leverage at one half; an interior maximum of the growth exponent at or below two; a ratio pinned at one by construction that the field says will move. None is trivially true and each can fail.

H3. It cannot be shown wrong. Rejected. Four dissolution results are printed in Section 14, each tied to a preregistered study, with hard kills and rival-favouring nulls, and the field's name carries its own death clause.

H4. Policy relevance is premature; talking about regulators is safety-washing. Conceded on timing. Policy weight follows results, never formulation; Section 13 is written in the conditional for that reason.

H5. Even if all five laws hold in language models, that is AI engineering, not a field. Conceded as the demanding bar, and it is the right bar. A field needs the substrate crossing: the same exponent structure in at least one non-artificial system. That is why the fourth-cell test is part of the founding conjunction in Section 11.1 and not an optional extra.

I. What is not answered here

I1. There may be objections this list has missed. Conceded, by construction. The list is published so that it can be extended, and every objection added from outside will be printed with its disposition in the next version, under the standing invitation that ends this paper.

13 What this buys, if it holds

Fields earn adoption when strangers can compute with them. If the laws survive their deciders, the following become calculations rather than debates: given a measured γ for a corrector class, the maximum sustainable growth exponent of any system it oversees follows from the ARC Ceiling; given measured drift, the required correction rate follows from Law II; oversight architectures acquire a specification sheet (the corrector's composition class, its measured γ, its implied ceiling) the way engines acquired efficiency ratings; and the safety question "can oversight keep up" becomes a comparison of γ measured for each corrector class rather than a scenario argument. Everything in this paragraph is conditional on results that do not yet exist, and is written in the conditional for that reason.

13.1 And if it fails

The negative result is also a contribution, and it is one the oversight programme needs. A ratio of 1.00 holding tight everywhere licenses same-class ladders; a universal absence of any regime where correction out-scales drift says self-correction never wins, which is a different and worse world that policy should know about. The designs commit to publishing either outcome under the same title. A field whose failure is informative is not a vanity project.

14 What would dissolve the field

If the correction-versus-drift race finds no regime where correction out-scales drift, Law II has no subject matter. If the corrector-class contrast runs the other way, or falls wholly below the preregistered support threshold wherever the cap should bind, the field's distinctive mechanism is dead and its remains return to the hosts named in the dissolution map of Section 3.1. The trial's own operating characteristics decide which observations are available: on a roster whose targets genuinely differ it returns an inconclusive verdict rather than an affirmative equivalence, so a null is a failed attempt to confirm and the dissolution reading rests on the directional and below-threshold outcomes named here. If the titration finds no interior maximum and no ceiling behaviour, the central impossibility dissolves. If the cross-domain families fail the fourth-cell test, substrate independence was an artefact of family flexibility. Any one death wounds the field, and this paper commits to publishing each under this title.

A name given before its tests must be able to die with them, and the record must be able to say which death it was. A law failing in its stated form kills that law and the theory that stated it; the field carries

the corrected form, or none, as the record allows, and a reader who rejects all five laws and keeps the measurement is still inside it, by the minimal commitment of Section 15.2. The name itself dies when the object goes: if the corrector-class contrast fails in either way named above, the residue no host could state has proved empty, every piece returns to its host, and this paper will say so under this title rather than quietly disappear. If the laws survive their registered tests and hold outside the domains that raised them, the name has earned its generality. Either way the record decides, not the naming.

15 What this paper does and does not show

The paper founds the field: it sets out the object, the variables, the laws as named conjectures under test, one bound derived inside a stated model, and the instruments and deciders for Laws I to III. It confirms no law, and none of its decisive experiments has run, because none runs before its registration is public. What would have to happen before the field could be called a science is set out first, and what the field is not, with the minimal commitment, second.

15.1 What would make it a science

Naming a field does not make it a science. The entry criteria Recursive Dynamics has not yet met, in public:

Table 5. The entry criteria Recursive Dynamics has not yet met.

Criterion	Status	What would meet it
Registered prediction success	Not met	One decisive experiment returning its predicted direction under its registered analysis
Independent replication	Not met, as of 24 October 2026	The same result from hands that are not the author's; every instrument, seed, schedule and threshold is written down for this reason, and each study's design becomes public with its registration
Use by strangers	Not met	The state variables appearing in other people's work because they needed them
Survived audit	In progress	The record of public retractions, minute-resolution self-demotions, adversarial cold reads and printed limitations, offered as the standard future claims accept

15.2 What the field is not, and the minimal commitment

The field is not a forecast of when machines surpass people, not a theory of intelligence, not an ethics, not a theory of everything, and not a claim about consciousness. Its laws switch on where a corrector exists inside a growing system and switch off where none does; the composed hierarchies that its scaling mathematics bears on (stars included) are graded as correspondence of form and no further. The programme's speculative coda on recursion at cosmological scale is separable and sits outside the field; a reader can reject it entirely and lose nothing here.

The minimal commitment required to work in the field is small and stated so that it can be refused: that the self-improvement loop is a legitimate object of measurement, with quantities that can be defined, instruments that can measure them, and relations that can be shown wrong. A reader who accepts that and rejects all five laws is inside the field; a reader who accepts the laws and refuses measurement is not.

16 Limitations

The paper prints its limitations where they arise; they are gathered here plainly, each with the place it is argued.

- The correction leverage γ has never been measured, by anyone, so the value two rests on a premise the programme names as the conjecture's most likely point of failure (Section 6; objections B4 and G2).
- The bound of Section 6 is a conditional crossover inside a minimal model: it does not establish bounded harm, finite-horizon safety or a law of nature, and its treatment of depth has been criticised in review (objection B5).
- When models score models, the scorer can share the blind spots of the scored; the circularity is reduced, disclosed and measured, not abolished (Section 10; objection C4).
- Whether five state variables suffice, and whether the quantities the studies measure are the objects the laws name, are open questions (objections C1 and C2); the capability ladder is calibrated but not yet validated by other hands (objection C5).
- The minimal model assumes stationarity and separability of drift and correction, and reads capability on one scalar axis (objections G3 and G4).
- Substrate independence stands at exploratory grade, and the fourth-cell test that would catch flexible fitting has not run (Section 9; objection C3).
- The decisive experiments are preregistered by timestamp and not run, and the field has one author, no replication by other hands and no use by strangers (Sections 11 and 15.1; objections B1, D2 and D3).

17 Conclusion

Until the entry criteria are met, one sentence carries the field's whole status: **Recursive Dynamics is a new field at the Carnot stage: one memoir in, instruments built, decisive measurements preregistered by timestamp and not yet run, staked in public where failure will be as visible as success.**

The invitation that follows from everything above is the only one the field can honestly extend: run the test that could kill it. Every instrument, seed, schedule and threshold is written down for that purpose. The designs of the decisive trials are lodged privately on OSF today, and each becomes public with its registration. An independent run outranks the programme's own.

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A Version history and corrections

Every version and correction, with its date. Entries from earlier versions are kept as printed; the section and figure numbers they name are those of the earlier layout, which Table 6 maps to this edition.

A.1 Version 3.5

Version 3.5, 24 October 2026. The paper is set in the form of a leading arXiv paper: a title in sentence case taken from the paper's own question, a secondary line and a first-page footnote that state its place in the programme, the abstract on the first page, a contents page, a list of contributions and a map of the paper in Section 1, the debts of the former section 13 as the related-work section (Section 2), with every work cited at its official source and with the earlier formal treatments of recursive self-improvement and recent work on oversight added, the former sections 15 and 16 under the heading "What this paper does and does not show" (Section 15), a Limitations section that gathers the limitations the paper prints (Section 16), the closing section headed Conclusion, and lettered appendices. Laws IV and V are stated as registered as laws on 15 October 2026 in version 1.103 of the theory registration and not yet tested, their mathematics in development, in the abstract and in Section 5. Unit 06 is stated as registered on OSF on 1 October 2026, public from 4 October 2026, run on 6 October 2026 and reported as administratively incomplete, with no inferential verdict, and registered is said of the theory registration with its updates and of unit 06 (Section 11). Law II's line now gives the law in the registration's words, without the reading of necessity that the next sentence already says is not established (Section 5). Section 6 no longer says that past the crossover the loop chokes on its own errors, which goes beyond the four lines; it says that the value one half for γ is a premise and not a consequence of independence; and it says that growth up to quadratic in depth is the most a same-class corrector can sustain, reached only at the ceiling, where the tie decides nothing by itself. Proposition 1 carries its one-line proof. Section 4 says that capability and recursive depth are states and that the correction rate, drift and correction leverage are elasticities of relations among them, and its table reads drift as an exponent. Section 5 adds that each of the three statements of one boundary is read in its own model. The commitments printed under the abstract are folded into Section 1, where their prohibitions already stood. The figures are redrawn as vector figures, their words brought to the text (five laws, each a named conjecture; designs preregistered by timestamp; no claim that the loop chokes); the scorecard figure is not redrawn, because Table 4 carries it. Every mention of one of the author's papers links its OSF DOI. Planned extensions not yet preregistered are now called set out for registration.

Section 1 gives the why and the result in plain words; it no longer says that the literature answers what bounds these systems with scenarios and intuitions alone, since that literature also has formal models, bounds and fitted scaling laws, each in its own variables; and it says that cybernetics was proposed before most of its results existed, where version 3.4 said before its results existed. Prior work is conceded exactly what it states: to cybernetics, the requisite-variety intuition and the good-regulator theorem (Table 1, Section 2 and objection A1); to the oversight literature, its bounds, where version 3.4 said it had no bound on what a corrector can do (Table 1, host 7, the residue, Sections 8 and 13, objection A4); to the literature on models trained on their own outputs, the correction rate it pairs with drift, where version 3.4 said it measured drift without one (host 5 and the residue); the self-modifying-program line and the framework Section 8 calls the strongest are named; and each host of Section 3.1 that rests on a canonical work cites it. Objection A5 is partly conceded where it was a reading debt, because that literature pairs drift with a correction rate, as its own condition foresaw, and the count of Section 12 becomes thirty-nine conceded or partly conceded and one handed to experiment, where it was thirty-eight and two.

Laws IV and V are restated in the two-limb form that version 1.103 of the theory registration registers (Section 5): Law IV's retention limb is P15 and its comparison limb has no registered proposition yet,

where version 3.4 said of that comparison that it "has no registered refuter yet"; Law V's build-order limb carries the direction that version 1.103 fixes beside P12, which keeps no direction, and the rule that joins the limbs replaces version 3.4's sentence "How the two verdicts make one verdict on the law is not fixed yet". The mathematics of Laws IV and V is cited at its current version, 1.1 of 7 October 2026, and is said to be in development, as its own status paragraph says that fifteen questions stay open. Law V's deciding study is stated as designed and awaiting registration; the record's snapshot is that of 4 October 2026, with the tests already run counted beside the designs; replication and use by strangers are stated as not met as of this version's date; and the lead of Section 15 gives the reason no decisive experiment has run.

The abstract says that self-improvement cannot stably outrun its own correction, where version 3.4 said cannot outrun, which read against Law II as a guarantee. The criticism of the derivation's treatment of depth is said to have been raised in review, where version 3.4 attributed it to an outside reader and spoke of published criticism that it did not cite. Objection B2 is partly conceded where it was conceded: each law is a law under test, and that Laws II and III are proved only inside minimal models stays conceded. Objection D3's heading says no doctorate, where version 3.4 said no credential. Section 4 says that thermodynamics could not state its laws in their quantitative form until temperature was absolute, since Carnot's bound of 1824 came before the absolute scale of 1848. Section 9 counts four temporal checks as matched, where version 3.4 said confirmed beside the sentence that none of this is confirmation. No sentence sends the reader to a register without a DOI, and Appendix D.3 no longer says that the registers govern the text; Appendix D.1 says registered, not preregistered, of the theory registration. Law I's symbol I is defined, Laws II and III carry status lines, and the dissolution map gives Laws IV and V their registration. Smaller corrections: Section 12 no longer opens with a version number and says why group D has no D4; Section 14 states its dissolution condition once; two terms of the instruments are defined where they are first used and four more in the glossary; capitals used for emphasis are set in italics; and the Conclusion no longer says that its status sentence appears wherever the field's name does. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

A.2 Version history

1. v1.0, 28 August 2026, first draft.
2. v1.1, same day, adding the worked derivation of the Ceiling, the measurement-scale founding principle, the steelmanned rival with the programme's contrary pilot, the what-this-buys section, and the registered numbers of the decisive experiments.
3. v1.2, 29 August 2026, the capability ladder restated in the registers' canonical forms, three further neighbouring fields named with what each owns, historical precedents held to the well-attested set.
4. v1.3, same day, the state variables written exactly as the notation register defines them (the correction rate β_C replaces a bare reinvestment β; reinvestment named as the experimental manipulation).
5. v2.0, same day, the case for the field scored against six licence conditions, forty-two objections collected and dispositioned, the scorer-circularity limitation printed, what the field is not and the minimal commitment, and the entry criteria as a scored table.
6. v2.1, same day, the precedent class of limit-theorem fields (6.1), the founding conjunction with the cross-instrument consistency check and the registered-versus-proposed separation (10.1), what the field buys if it fails (12.1), and eight further objections (A10, G8, G9, I1 to I5) with dispositions.
7. v2.2, same day, the forbidden outcomes stated up front in the abstract and section 1.1 (an outside reader's one reservation, that the concessions could read as silence), and five figures: the loop, the crossover, the scorecard, the ratio zones, the founding conjunction.
8. v2.3, same day, section 2.1: the honest attempt to house the questions in twelve neighbouring fields,

the concessions that attempt forces, the dissolution map by host, the residue no host holds; objection A11. The figures were redrawn the same evening.

9. v2.4, 30 August 2026: the retired $\alpha_{\max}$ form in section 4 replaced by the notation register's α_{crit}; the scorecard figure's condition 1 evidence line aligned with section 2.1 (twelve hosts tried); the crossover figure's label written α_{crit}.
10. v2.5, 30 August 2026: Law II carries its canonical name, the ARC Co-Scaling Law, and Law III is written the ARC Ceiling at every mention in the body (the bare forms were retired across the programme on 30 August 2026; this history keeps its own earlier wording); the objection groups run A to I without a gap (the former group I is now H, the former J is now I) so that the paper and the field's public page cross-reference by the same letters, and the earlier history entries keep the letters current when they were written. No claim, status or disposition changed.
11. v2.6, 30 August 2026: one further objection in group A, raised by the author against his own field (that Recursive Dynamics is the ARC Theory renamed), rejected with a separability test; fifty-two objections, eleven rejected. The name's kill condition in section 14 corrected: the earlier wording had the name die with the laws, which contradicted the minimal commitment of section 16 (a reader who rejects all three laws and keeps the measurement is inside the field) and the separation of questions from answers; the name now dies with its object, the class-set cap on correction coupled to growth in depth, and the record says which death it was. D1 aligned with the corrected clause. The field's public page carries the same objection and clause, with a decision figure of the kill condition, from the same date. v2.7: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed. v2.8: The remaining AI-assistance statements have been removed.
12. v2.9, 18 September 2026: two figures corrected to match this paper's own text. Figure 4, the ratio zones, states four verdicts where it had stated three, so that a true null inside the band returns an inconclusive verdict and never a refutation, and the caption carries the same sentence; figure 2, the exponent balance, now closes by saying neither has been run. No claim, status or date has changed.
13. v3.0, 3 October 2026: the theory is stated with five laws. Laws IV and V, added on 24 September 2026, enter section 4 beside Laws I to III, each with its status (its mathematics in development; neither registered as a law nor tested) and the registered propositions that carry it, with one paragraph on what their mathematics proves and does not. Law I's status now prints the measured row whole, each uncertainty with its own estimator, as the theory registration requires: the interval that includes zero and two belongs to the endpoint estimate of 0.59, and the regression estimate near 0.49 carries a standard error of 0.20. The sentences that counted three laws now count five, or name Laws I to III where those three were meant. No registered law, prediction or kill condition changes; no result is added; no date of any claim moves.
14. v3.1, 3 October 2026: the mathematics of Laws IV and V is cited at its version 1.0, frozen on 3 October 2026, and is no longer called a draft or said to be in development; each of the two new laws gains its short form (the retained fraction below one half; the registered minimum effect of build order). The study designs, which earlier versions called registered, are called preregistered, short for preregistered by timestamp, and registered is kept for what OSF has accepted. Section 10 gives the standing of the trials as read on 3 October 2026, where version 3.0 said that all were awaiting submission and that nothing had been submitted. The objections are counted from the printed list, which runs D1, D2, D3 and D5 in group D: fifty-one, of which thirty-eight are conceded or partly conceded, eleven rejected and two handed on. The abstract had fifty-two and thirty-nine, and section 11 had thirty-nine and ten. Section 1.1 and objection G7 give the forbidden outcomes as the abstract gives them. The scorecard counts five laws and says which of its rows cover Laws I to III

alone; it no longer prints a bare zero, and nor does section 6.1. Host 3 of section 2.1 names Laws II and III where it spoke of two other laws, and the lead to the boundary table in section 4 no longer calls its three statements Laws I to III. Section 7 prints the measured row whole and corrects five of six models to four of the five models measured, as Paper II reports. The status of Law I names the register that tests it, and section 10 says where the theory registration places the titration's proposition. The dissolution map says that Laws IV and V are not yet placed on it, and the abstract and section 14 follow the map. Objection D5 and the line under the title date the phrase recursive dynamics from the sealed record of 8 December 2024. The closing and the entry criteria no longer say that every instrument, seed, schedule and threshold is published: the designs of the decisive trials are lodged privately on OSF and become public with registration. Smaller corrections: a pointer to objection I5 now reads H5; Law III is written the ARC Ceiling at three mentions that had the bare form; objection G6 records that the thirteen-minute demotion was later reversed; objection A1 points to objection A11 where it had pointed to section 14; objections A5 and D3 and host 10 of section 2.1 claim no more than has been read or linked; and the entries of this history for versions 2.3, 2.5 and 2.9 are reworded in plain terms, the first of them no longer describing how the figures were redrawn. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

15. Version 3.2 (4 October 2026). The abstract is shortened to the paper's claim and its status; the commitments the longer abstract carried now stand beneath it under their own heading, in the same words. Figures are numbered and keywords added. The PDF is typeset afresh. No claim, number, status or reference has changed.
16. Version 3.3 (4 October 2026). Retitled from "Recursive Dynamics: The Proposal of a Field" to "Recursive Dynamics: Founding the Science of Systems That Improve Themselves", at the same DOI, and the line under the title now carries the new title's words; the commitments made to publish under the old title hold under the new one. Wording that presented the author as proposing the field he founded, or this paper as a proposal, now says founding, and a variable, relation or bound put forward for testing is now named, set out, added, conjectured or put to the test. Every statement of the field's stage keeps its facts; the closing sentence reads "a new field at the Carnot stage". Where proposed describes other people's work or extensions not yet preregistered, it is unchanged. No claim, number, status or reference has changed. The byline names the author founder of Recursive Dynamics.
17. Version 3.4 (5 October 2026). The title now follows the programme's uniform order (paper, its law or framework, its own title), *Founding Paper · Recursive Dynamics · The Science of Systems That Improve Themselves*, and the heading carries the whole title; until this version it read *Recursive Dynamics: Founding the Science of Systems That Improve Themselves*, under the heading *Recursive Dynamics*. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

A.3 Where the earlier layout now stands

Table 6. Where each part of the earlier layout stands in this edition.

Earlier layout (version 3.4)	Where it stands in this edition
Title block: title, theory-position line, line under the title, byline, framework dated record, authorship line	Title, secondary line, author block and first-page footnote; the dated record in Appendix B.2 and Appendix D.3; the authorship line in the copyright line

Earlier layout (version 3.4)	Where it stands in this edition
Abstract and keywords	Abstract; keywords in the document metadata
What the paper commits to	Section 1: the contributions list and Section 1.1
1, 1.1	Sections 1 and 1.1
2, 2.1	Sections 3 and 3.1
3	Section 4
4	Section 5
5	Section 6
6, 6.1	Sections 7 and 7.1; Figure 3 of earlier versions is carried by Table 4
7	Section 8
8	Section 9
9	Section 10
10, 10.1	Sections 11 and 11.1
11, groups A to I	Section 12
12, 12.1	Sections 13 and 13.1
13	Section 2
14	Section 14
15	Section 15.1
16	Section 15.2
(none)	Section 16
17, Closing	Section 17, Conclusion
Version history	Appendix A.2
Figures 1, 2, 4 and 5	Figures 1, 2, 3 and 4

B Priority record

Each dated record bearing on this paper and its field, with its kind of timestamp (Table 7), and the framework's dated record as earlier versions printed it under the title.

B.1 Dated records

Table 7. Dated records, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
8 December 2024	The phrase recursive dynamics in the author's sealed record: the book draft, line 1180 (also line 1622), and Version 2, line 1079	Book draft and Version 2, OSF 10.17605/OSF.IO/QASRK	Mail-server date, held privately; OSF deposit 5 October 2026

Date	Record	Identifier	Kind of timestamp
2 January 2026	The framework first public, in print: the first edition of the book (Eastwood 2026a)	ISBN 978-1806056200; OSF dqm3e	Publication date (author's record; the interior reads 6 January 2026); OSF deposit 26 August 2026
12 August 2026	The first of the programme's designs preregistered by timestamp	Bitcoin block 962066	Bitcoin anchor
16 August 2026	Law III's corrected form, $\alpha_{\text{crit}} = 1/(1-\gamma)$; the retired form, with γ alone in the denominator, retracted	Statement paper, OSF file, revision 27	OSF deposit
27 August 2026	The phrase adopted as the name of the field (priority register PC-RD-001)	Stated in this paper (version 2.6, OSF deposit 31 August 2026)	Date written inside the document
28 August 2026	This paper's first draft, version 1.0 (PC-RD-002)	Appendix A.2, entry 1	Date written inside the document
30 August 2026	Version 2.6: the name's kill condition corrected, and the objection that the field is the ARC Theory renamed added (PC-RD-003)	OSF file, component hcpbu	Date written inside the document; OSF deposit 31 August 2026
31 August 2026	This paper first published on OSF	DOI 10.17605/OSF.IO/HCPBU	OSF deposit
8 September 2026	The theory registration, twenty-two propositions, version 1.100; updated 13 September 2026 (version 1.102)	DOI 10.17605/OSF.IO/P8CKQ	Registration accepted on OSF
24 September 2026	Laws IV and V added to the theory	This paper from version 3.0 (Appendix A.2, entry 13)	Date written inside the documents
3 October 2026	Laws IV and V: the mathematics, version 1.0	DOI 10.17605/OSF.IO/4B3VT	Date written inside the document; OSF deposit 3 October 2026
7 October 2026	Laws IV and V: the mathematics, version 1.1	DOI 10.17605/OSF.IO/4B3VT	Date written inside the document; OSF deposit 12 October 2026
15 October 2026	Version 1.103 of the theory registration, which registers Laws IV and V as laws	DOI 10.17605/OSF.IO/P8CKQ	Registration update approved on OSF

B.2 The framework's dated record

Framework dated record: 8 December 2024 (private, sealed); 2 January 2026 (first public, in print). The phrase recursive dynamics stands in the sealed record of 8 December 2024; it was adopted as the name of the field on 27 August 2026.

C Notation and glossary

The symbols the paper uses, each with the place it is defined (Table 8), and its terms in plain language (Table 9).

C.1 Symbols

Table 8. Symbols, with the place each is defined.

Symbol	Meaning	Defined in
U (or C)	capability: what the system can do, on a calibrated ladder with no ceiling	Section 4
R	recursive depth: how many times outputs have re-entered the loop	Section 4
β_C	correction rate: how fast internal correction strengthens as capability grows	Section 4
k	drift: how fast the errors and misalignments that accumulate per self-modification grow as capability grows	Section 4
γ	correction leverage: how correction capacity scales with capability; the exponent in the ARC Ceiling	Sections 4 and 6
α	the growth exponent: capability grows as a power of recursive depth	Sections 5 and 6
I	base capability: what the system can do before any self-improvement, the same battery score read at recursive depth one (the notation register's base intelligence); the multiplier in Law I, $U = I \times R^{\alpha}$	Section 5
α_{crit}	the crossover exponent, $1/(1-\gamma)$: Law III, the ARC Ceiling	Section 6, equation (2)
C_0, R_0	the reference capability and depth of the growth law $C(R) = C_0(R/R_0)^{\alpha}$	Section 6
$B(R)$, b	correctable burden per unit depth, $b\,\mathrm{d}C/\mathrm{d}R$, and its coefficient	Assumption 1
$A(R)$, a	correction service capacity, $a\,C(R)^{\gamma}$, and its coefficient	Assumption 2
N	the number of corrections a corrector aggregates	Section 6
$S_{\text{ext}}(1)$	the retained fraction of an externally installed gain at the matched budget (P15)	Section 5
γ_{first}, γ_{after}	the correction-leverage exponent with correction formed first and with correction added after (P12)	Section 5
δ	how fast a loop compresses its own cycle time (Paper XIII)	Section 5, Table 3
P1 to P22	the propositions of the theory registration	Sections 5 and 11
ARC-1	the register of the theory registration that tests Law I	Section 5
p	a binomial p-value	Section 9

C.2 Plain-language glossary

Table 9. The paper's terms in plain language.

Term	In plain language	See
Recursive system	Any system whose outputs re-enter its own improvement process	Definition 1
Native object	The self-improvement loop, which no neighbouring field takes as primitive	Section 3
Scale discipline	Capability measured on a ratio scale with a true zero and no ceiling, before any law is stated	Section 4
Capability ladder	A latent, ratio-scaled capability measure with a true zero and no ceiling, on which doubling means the same thing at every height	Section 4
Corrector class	What a corrector is made of; a same-class corrector shares the architecture, training distribution and blind spots of the system it corrects	Section 6
Independence-aggregation benchmark	The premise that caps a same-class corrector's correction leverage at one half	Sections 5 and 6
Correction-versus-drift race	The critical-correction-ratio study, which decides Law II	Section 11
Eclipse measurement	The corrector-class ratio, which decides Law III's mechanism	Section 11
Capability-ladder titration	The ARC Bound study, which decides the ceiling's behavioural signature	Section 11
Fourth-cell test	The test on the logarithmic family and on domains never fitted, which exists to catch the programme flattering itself	Section 9
Laundering	Re-presenting every output so that no scorer can tell what produced it	Section 10
Fixture-gated estimator	Estimator code whose every preregistered verdict branch must pass a planted-truth test before anything ships	Section 10
Defect clause	The preregistered rule under which the specification, not the code, is the analysis, so a coding error found after registration is a disclosed repair	Section 10
Forced null	The corrector-class ratio fixed at exactly 1.00 by construction if what a corrector is made of does not matter	Section 11
Founding conjunction	The results no neighbouring field could state in its own variables, taken together	Section 11.1
Dissolution map	Where each part of the field goes if the founding conjunction fails	Section 3.1
Licence conditions	The six checkable conditions by which a new area of science is licensed	Section 7
Preregistered	Fixed in one of the dated designs, short for preregistered by timestamp; registered is kept for what OSF has accepted	Section 11

Term	In plain language	See
Reading debt	A literature that must be read before a point can be argued either way, under the rule that nothing is cited before it is read	Section 12
Minimal commitment	That the self-improvement loop is a legitimate object of measurement, with quantities that can be defined, instruments that can measure them, and relations that can be shown wrong	Section 15.2
Carnot stage	One memoir in, instruments built, decisive measurements preregistered by timestamp and not yet run	Section 17

D The paper's place in the programme

The theory registration this paper refers to, the paper's place among the laws of the ARC Theory, and the registers its claims are tied to.

D.1 The theory registration

Theory-level predictions registered on OSF: DOI 10.17605/OSF.IO/P8CKQ, registered 8 September 2026 (version 1.100), updated 13 September 2026 (version 1.102) and 15 October 2026 (version 1.103), twenty-two propositions. Companion: Register · The ARC Theory · Operational Definitions, version 1.9.1, 5 October 2026.

D.2 Placement

This is the founding paper of Recursive Dynamics, the field as a whole. It states the field's object, its measurement principle and its five state variables (Sections 3 and 4), and it states the five laws of the ARC Theory, the field's founding theory, as named conjectures under test (Section 5): Law I, the ARC Principle, Law II, the ARC Co-Scaling Law, and Law III, the ARC Ceiling, registered on 8 September 2026; Law IV, the ARC Persistence Law, and Law V, the ARC Embedding Law, added on 24 September 2026 and registered as laws on 15 October 2026 in version 1.103, neither yet tested, their mathematics in development. It derives Law III inside a stated model (Section 6). It decides no law and makes no measurement of its own; its decisive experiments are preregistered by timestamp and not run (Section 11). It does not treat the Eden Protocol or the HARI Treaty.

D.3 The registers behind the claims

Every claim in this paper is tied to the programme's registers of claims, corrections, kill conditions and antecedents, which carry per-claim statuses; where a register and this text differ, the text is corrected at the paper's next version and the correction listed in Appendix A.